

Victor Chua

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Singaporean Citizen | [linkedin.com/in/vicchua123](https://www.linkedin.com/in/vicchua123) | Available part-time, 15–20 hrs/week from August 2026
AI/LLM Solutions | On-Premises & Localised Deployment | Evaluation & Internal Enablement

Education

Nanyang Technological University (NTU) Aug 2024 – Dec 2028
Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

Work Experience

SUSS Digital Solutions & AI Singapore
AI & Software Engineer Intern May 2026 – Aug 2026
On-Premises LLM Document Automation & Evaluation

- Built an on-premises pipeline on a locally hosted **open-source Llama 3 3B**, so sensitive invoice data never left the network; a **human approval gate** and digital signature precede send, cutting 10 minutes to one click
- Designed an **evaluation harness** for testing **LLM reliability** on lecture diagnostics: faster-whisper transcription into LLM-scored rubrics, plus a composite metric ranking delivery quality cohort-wide
- Automated lecture quality review with **FastAPI** over the **TwelveLabs** API, which judges engagement semantically rather than detecting silence, returning trimmed footage plus flagged timestamps from 3-hour recordings
- Co-presented AI tool evaluations and cost-benefit analyses to **non-technical senior leadership**; translated technical deployment decisions into business terms and addressed reliability concerns

Yap Motor Singapore
Machine Learning Intern Dec 2025 – Jan 2026
Production Retrieval System for an Internal Knowledge Base

- Built Yap Motor's first AI system, a production **RAG** pipeline (Google Gemini with **ChromaDB**) letting 24 mechanics query 500+ pages of repair manuals in real time; still in daily use six months later
- Designed the retrieval **data flow** in four stages: folder priority scoring, **regex** extraction of vehicle make/model/year from natural language, dynamic ChromaDB filters, then re-ranking before the LLM
- Reached **90% retrieval precision** across 8 diagnostic categories by tuning chunk size, overlap, and embedding strategy, validated over a month of mechanic feedback on whether results answered the query

DBS Singapore
Data Engineer Intern May 2023 – May 2024
Enterprise Data Pipeline Monitoring & Integration

- Instrumented a **Grafana** and **SQL** dashboard monitoring a **3,000+** daily job production pipeline, surfacing version-specific bottlenecks that the owning teams then fixed, resulting in **20% less RAM** and **15% faster runs**
- Developed an **Elasticsearch REST** client and integrated it into the bank's internal Python library, giving the team programmatic real-time feature retrieval for production model inference for the first time

Research Projects

Human-in-the-Loop LLM Translation Interface Singapore
Research Assistant, NTU Turing AI Scholars Programme (TAISP) May 2025 – Aug 2025

- Implemented **bidirectional generation** for a computer-assisted translation prototype: forward source-to-target suggestions, and backward refinement where an edit updates preceding suggestions in real time
- Co-designed the **search-based approach** behind it, replacing single-suggestion machine translation with multi-option, translator-in-the-loop generation; presented as a poster supervised by Dr. Grandee Lee

Personal Projects

Drum Hub – AI drum-score reader | hf.co/Icyautumn/drum-omr-mobilenetv3 Jun 2026 – Present

- Developed a **dual-head MobileNetV3** OMR model via two-phase transfer learning, reaching **90.9% duration accuracy** and 0.93/0.89 F1 on kick and snare over 1,924 bars from 26 unseen songs
- Published the model and **ONNX**, the score reader runs offline on a student's machine with no network call and no per-use API cost

MusicBase – Music education platform for teachers, students and parents Jul – Aug 2022

- Shipped a cross-platform **Flutter/Firebase** app to iOS, Android and web in two months, with **role-based access control** giving teacher, student and parent accounts distinct interfaces and data visibility. reached 10 active teachers and 50 active students at peak

Leadership & Activities

TAISP x MLDA Workshop Lead, NTU School of Computing 2025
Designed and led an **internal training** workshop on Linear Regression and Neural Networks for **207 participants**; built all slides, annotated Jupyter notebooks, and graded exercises from scratch, with a 100% completion rate

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Bash
AI/LLM: RAG, ChromaDB, Semantic Chunking, Re-ranking, On-Premises Inference (Llama 3 3B), LLM APIs (Gemini, GPT, Claude), Prompt Engineering, LLM-as-Judge Evaluation, PyTorch, HuggingFace, ONNX Runtime
Solution Architecture & Security: Data Flow Design, On-Premises Deployment, Role-Based Access Control, Human-in-the-Loop Approval Gates, REST API Design, FastAPI, Elasticsearch Integration, Multi-Stage Retrieval Pipelines
Documentation & Enablement: Workshop Curriculum, Teaching Notebooks, Non-Technical Stakeholder Briefings